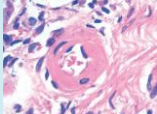
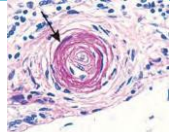
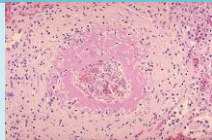
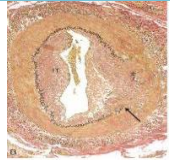
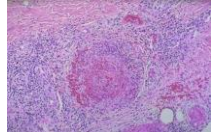
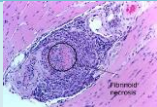
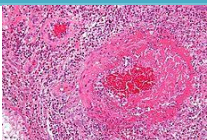
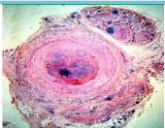
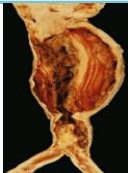
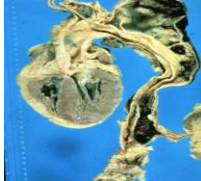
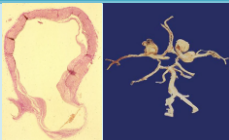
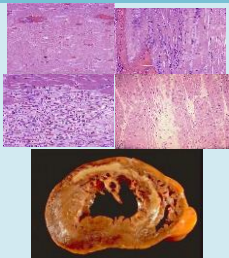
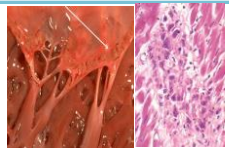
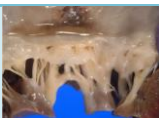
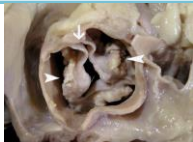
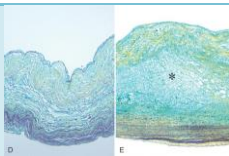
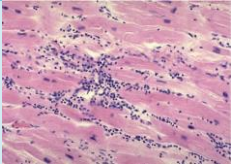
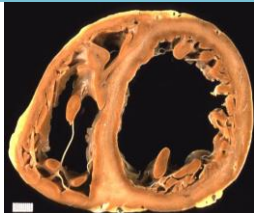
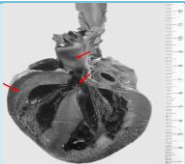
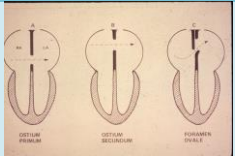
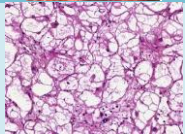
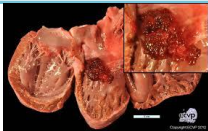


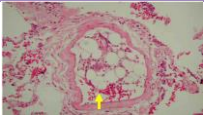
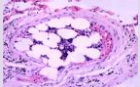
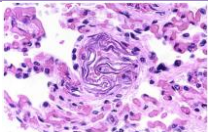
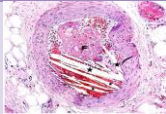
Disease (CVS)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Atherosclerosis		-Male 40-60 yo -Asymptomatic -Hypercholesterolemia -Hypertension -Smoking -Diabetes	-Occlusion -Rupture	-Intimal disease -Lumen narrowing	-Infra-renal abdominal aorta > proximal coronal artery > branches of abdominal aorta > internal carotid > circle of willis.
Monkenberg's medial calcific sclerosis		-Elderly patients -Incidental finding -Asymptomatic	-None	-Lumen stays same -Dystrophic calcification -Tunica media	-Muscular arteries: radial and tibial.
Arteriolosclerosis (Hyaline)		-Chronic benign HTN -Asymptomatic -Incidental finding -Diabetes	-Renal failure -Arteriolonephrosclerosis -Chronic renal failure	-Pink amorphous app. -Lumen narrowing -Intima/media	-Affects afferent arteriole ONLY!
Arteriolosclerosis (Hyperplastic)		-Chronic malignant HTN -BP: >200/120	-Acute renal failure -Basal ganglia hemorrhage -Chronic renal failure	-Concentric, laminated thickening of the walls -Onion skin -Lumen narrowing -Media	-Small vessel (arterioles).
Necrotizing arteriolitis		-Sudden malignant HTN -Pheochromocytoma -Cocaine use	-Aortic dissection -Hemorrhagic stroke -Papilledema -Retinal hemorrhage -Acute renal failure	-Fibrinoid necrosis. -Flea bitten app. -Intima/media -Necrotizing arteriolitis	-Small vessel (arterioles).
Giant cell (temporal) arteritis		-Older than 50 yo -<u>Jaw claudication</u> -Facial pain -Polymyalgia rheumatica	-Blindness -Granulomatous vasculitis	-Intima (thicken)/inner 1/2 of media -Giant cells -IEL fragmentation -Elevated ESR -Skip like lesions	-Medium to large arteries: temporal, ophthalmic, and vertebral. -Type IV hypersensitivity.
Takayasu (Pulseness disease)		-Female <40 yo -Asian	-Aortic stenosis -Left heart failure	-Tree bark appearance -Elevated ESR	-Medium to large arteries with relative sparing below chest.
Polyarteritis nodosa (PAN)		-Younger people -Hepatitis B -Foot/wrist drop	-Acute tubular necrosis -Renal failure -MI -Liver/GI infarct -Gangrene	-Lung is spared -Segmental involvement -Strings of pearls -NO giant cells	-Small to medium arteries EXCEPT for the lung. Capillaries not affected (NO hematuria). -Renal>coronary>hepatic>GI. -Lesions of varying stages
Microscopic polyangiitis		-Hematuria -Hemoptysis	-Necrotizing vasculitis	-Elevated MPO-ANCA (used to be P-ANCA)	-Smaller vessels/capillaries. -Unlike PAN get lung involvement (hemoptysis) and hematuria! -Lesions in same stage

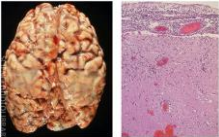
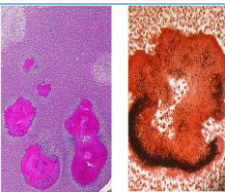
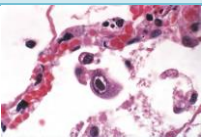
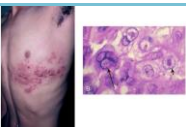
Disease (CVS)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Churg Strauss		-Asthma -Allergic rhinitis -Lung infiltrates	-Extravascular necrotizing granulomas. -Death (cardiac inv.)	-Eosinophilia -Elevated MPO-ANCA -Granulomas	-Small vessels: of cutaneous, GI, renal, and lung. -Differentiate from microscopic polyangiitis by necrotizing granulomas and eosinophilia!
Kawasaki syndrome (Mucocutaneous lymph node syndrome)		-Children <4 -Japanese -Asian -Rash on palms and soles -Strawberry red tongue	-Coronary aneurysm -MI -Cardiac tamponade	-Aspirin treatment -CRASH & BURN: Conjunctival injection, rash, adenopathy (cervical), strawberry tongue, hand/feet changes, fever (Burn)	-Small→Large arteries: coronary (Transmural w/ segmental fibrinoid necrosis). *Think kawasaki for rash/heart.
Thromboangiitis obliterans (Burger's disease)		-Males < 35 yo -Japanese, Indian, or Jewish descent -Smoker -6 Ps (pulselessness incl.)	-Gangrene -Auto-amputation	-Corkscrew app. on angiogram -Intermittent claudication -Reynaud's phenomenon -Neurovascular bundle affected -Microabscesses (PMN, CD4, thrombi)	-Small to medium arteries: tibial and radial→extends to veins and nerves. -Type IV hypersensitivity. -To tobacco.
Wegener's (Aka Wegener's) <i>Limited Wegener's doesn't involve the kidney</i>		-Males ~40+ yo -Hematuria -Hemoptysis -URT: Nasal ulcer/sinusitis -LRT: Cough w/ blood -Lung abscess	-Naso-mucosal ulceration -Necrotizing pneumonia -Renal failure -Iron deficiency anemia -Interstitial fibrosis -Congestive heart failure	-Crescent shaped glomerular nephritis -Neutrophil microabscesses -Elevated PR3-ANCA (aka C-ANCA) -Pauci-immune fluorescence -Giant cells	-Capillaries→medium arteries: naso-pharynx, lung, and kidney. -Has URT involvement unlike Goodpasture's syndrome. -Type IV hypersensitivity.
Abdominal (atherosclerotic) aortic aneurysm		-Males 40-60 yo -Pulsatile/expansile mass -Atherosclerosis -Smoker -Asymptomatic	-Rupture (>5cm) -Mural thrombus -Hypovolemic shock -Death -Hydronephrosis	-Intimal disease (pressures media) -True aneurysm (all 3 layers) -When >5 cm: -Flank pain -Hypotension	-Infra-renal abdominal aorta. -Moves from intima→media. -Pseudoaneurysm is when the adventitia still confines the lesion.
Thoracic aneurysm		-STI history (syphilis) - (+) FTA-Abs test -Pain radiating to back	-MI -Cor bovinum -Embolic stroke -CHF -Pulmonary edema	-Tertiary syphilis -Starts in vasa vasorum -Tree-bark app. -Obliterative end arteritis	-Thoracic aorta (abdominal too). -Moves from adventitia→media. -Blood negative for syphilis.
Aortic dissection		-Marfan syndrome –or– -Ehlers danlos -Hypertension (malignant) -Atherosclerosis -Pain radiates to back	-Ascending (backward): pericardial tamponade -Ascending (forward): stroke, MI, Renal failure, aortic regurgitation -Descending: Iliac/femoral - gangrene	-Double barrel aorta -Defective media -Widening of mediastinum -Tearing pain -Unequal pulse in upper ext.	-Aorta. -6 P's (pain, pallor, poikilothermic, pulselessness, paralysis, paresthesia).

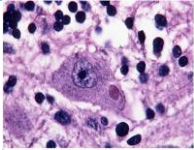
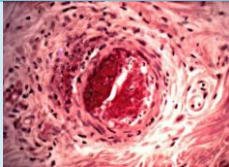
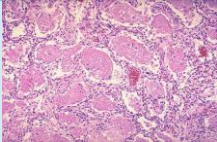
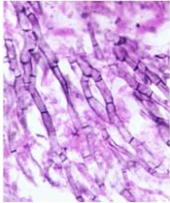
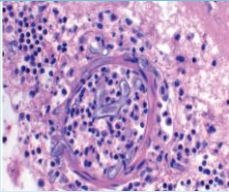
Disease (CVS)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Berry aneurysm		-Marfans syndrome –or- -Ehler danlos -Polycystic kidney disease -Asymptomatic	-Subarachnoid hemorrhage -Ischemic stroke -Communicating hydrocephalus	-Thunderclap headache -Saccular	-Between anterior cerebral and anterior communicating arteries.
Mycotic (infectious) aneurysm		-Microbial infection	-Endocarditis -Septic embolism	-Tricuspid→lung (cleared) -Mitral→brain	
Myocardial infarction <i>Sudden cardiac death is due to dysrhythmias, NOT infarction, so if revived see no EKG or enzymatic evidence of MI</i>		-Male >55 yo -Atherosclerosis -Hypertension -Diabetes -Hypercholesterolemia -Smoking -Obesity -Pansystolic murmur -LAD: L. lower sternal -RCA: Apex loudest	-1-24 hrs: cardiogenic shock or arrhythmia -1-3 days: fibrous pericarditis (transmural) -3-7 days: rupture of ventricular free wall (LCX), IV septum (LAD), or papillary muscle (RCA) -2 mth: Aneurysm or dresser's autoimmune	-Severe retro-sternal pain -Radiation of pain to left arm/jaw -Transmural-ST elevation -Atherosclerotic plaque -Subendocardial-ST depression -Hypovolemia -Elevated troponin I or CK-MB -Contraction band necrosis -Necrosis→inflammation	-Coronary artery block. -Fatal arrhythmias are V. fib and V. tach (#1 cause of pre-hospital death). -Ischemia to conduction system -NOT infarction! -Differentiate between MI and angina is MI <u>NOT</u> relieved by nitroglycerin (due to a block).
Acute rheumatic fever		-Children 10-15 yo -Repeated S. pyogenes inf. -Vegetation's NOT in chordae tendinae -Fever -Athralgia (no inflam.)	-Post streptococcal glomerulonephritis -Pericarditis -Mitral regurgitation -Left heart failure -Arrhythmias	-Aschoff bodies w/ anitshkow cells -Owl eye (T) or caterpillar (L) -Verrucae endocarditis -Elevated ASO and DNase B titer -M-protein molecular mimicry -Elevated ESR & CRP	-Mitral>aortic>tricuspid. -Type II hypersensitivity. -Jones criteria (joint, heart, nodules, erythema marginatum, Sydenham chorea. -No fibrosis.
Chronic rheumatic fever		-Age 25-30 yo -Recurrent attack of RHD -Murmur (both Sys & Dia) -Dysphagia/hoarseness	-Fibrosis of heart valves -Right heart failure -Nutmeg liver -Embolic stroke	-Commissure fusion -Chordae tendinae fusion -Fishmouth/buttonhow app. -Aschoff replaced by fibrous scars	-Mitral>aortic>tricuspid. -Differentiate because unlike normal wear and tear, fusion from calcification occurs!
Aortic stenosis		-Patients >60 yo -Congenital bicuspid-40 yo -Jaundice -SAD symptoms -Chest pain on exertion	-Left heart failure -MI -Embolic stroke -Arrhythmias -Hemolytic anemia	-Dystrophic calcification -Schistocytes (helmet cells) -Crescendo-decrescendo murmur -Presystolic click	-Aortic primarily. -Differentiate from CRF: -Fusion DOESN'T occur. -No fibrosis. -Aortic first (not mitral)
Mitral valve prolapse		-Women 20-40 yo -Marfans syndrome –or- -Ehlers danlos syndrome -Incidental finding -Polycystic kidney disease	-CT rupture -Mitral regurgitation -Left heart failure -Pulmonary edema	-Mid-systolic click -Pansystolic murmur at apex -Myxoid degeneration of valve -GAGs in spongiosa -Ballooning or hooding of valve	-Mitral valve. *Mitral valve prolapse and berry aneurysm associated with AD polycystic kidney disease.
Acute endocarditis <i>Janeway lesions, Osler nodes, and Roth spots</i>		-IV drug users -Sudden onset -High fever -Splinter hemorrhages	-Septic embolism -Arrhythmia (ring abscess) -Lung abscesses -Death	-S. aureus or fungi -Normal valves -Large vegetation's (septic) -Destructive to valves	-Tricuspid valve.

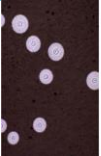
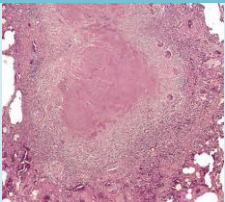
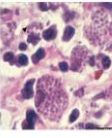
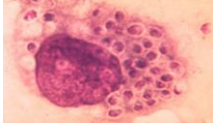
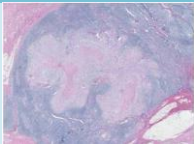
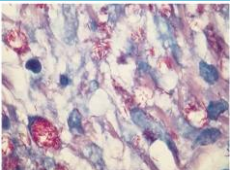
Disease (CVS)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Sub-acute endocarditis <i>Janeway lesions, Osler nodes, and Roth spots</i>		-Recent dental work -Low grade fever -Finger clubbing -Splenomegaly	-Septic embolism -Stroke -Glomerulonephritis -Healing	-HACEK group -S. viridans -Abnormal/damaged valves -Small vegetation's (septic)	-Mitral valve. -Type III hypersensitivity. -S. bovis is underlying colorectal issue (#1 COD is this case).
Prosthetic valve endocarditis		-Valve replacement -Fever -Murmur -Jaundice (metal valve)	-Microangiopathic hemolytic anemia -Stenosis -Insufficiency	-S. epidermidis most common -Gram (-) bacilli & fungi -Schistocytes (helmet cells)	-Mechanical valves the suture line and adjacent perivalvular tissue. -Bio-prosthetic valves the leaflets and perivalvular tissue.
Non-bacterial thrombotic endocarditis (Marantic)		-Hypercoagulable state -Endothelial abnormalities -Chronic illness/bed ridden -Adenocarcinoma (mucin)		-Sterile vegetation (fibrin, platelets, other blood components) -No inflammation/fibrosis -No bacteria	-Mitral valve.
Libman-sacks endocarditis (Verrucae)		-Systemic lupus erythromatosus -ANA Ab -Abortive miscarriages -Butterfly rash	-Mitral regurgitation	-Sterile vegetation -Inflammation	-Mitral valve with both sides affected.
Viral myocarditis <i>**Differentiate from MI because MI has necrosis before inflammation unlike viral myocarditis**</i>		-Flu like symptoms -Fever w/ ↑↑↑ heart rate -Myalgia's -Headache -Rigor -ECG: prolonged QRS -ST elevation -Cardiac enzymes (maybe)	-Arrhythmias -Dilated cardiomyopathy -Heart failure -MI	-Coxsackie B virus -Inflammation→necrosis -Patchy necrosis -CD8+ mediated damage -Plus direct viral damage	-Myocardium. -Complete resolution typically. -Others include: post-streptococcal glomerulonephritis, hemangioma, impetigo, lobar pneumonia, histoplasmosis. -Inhalation or fecal-oral trans.
Dilated cardiomyopathy		- <u>Dystrophin gene mutation</u> -Alcoholics -Viral myocarditis -S3 sound (gallop rhythm) -Displaced apex beat -Left and right heart failure	-Arrhythmia -Embolic stroke -Gangrene of lower limb	-Systolic dysfunction -Low ejection fraction -Dilated ventricles w/out increased workload or ischemic disease -Eccentric hypertrophy -Kentucky gallop rhythm	-All 4 chambers involved. -Cardiomyocytes elongate in response to volume overload with no disarray. -Associated with Duchene's.
Hypertrophic cardiomyopathy		-Sudden death -Athlete collapses & dies - <u>β-myosin heavy chain mutation</u>	-Arrhythmias -Right & left heart failure -Embolic stroke -Ischemia (IVS infarcts) -Infective endocarditis	-Diastolic dysfunction -Normal ejection fraction -Asymmetrical hypertrophy -Concentric hypertrophy -Banana-shaped left ventricle	-Left ventricular IV septum. -Cardiomyocytes increase in width in response to pressure overload with disarray.

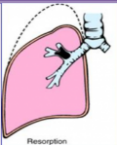
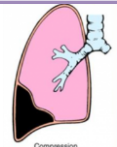
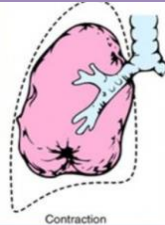
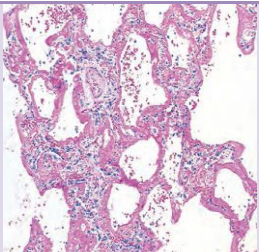
Disease (CVS)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Restrictive cardiomyopathy		-Low voltage ECG -Diminished QRS amp. -Hemochromatosis -Amyloidosis -Sarcoidosis -Loeffler's (eosinophilia)	-Arrhythmias -Heart failure	-Diastolic and systolic dysfunction -Decreased ejection fraction -Interstitial fibrosis	-Endocardium and myocardium.
Tetralogy of fallot		-Child >6 months -Tet spells (squatting) -Cyanotic spells -Finger clubbing	-DVT -Jet lesions -Right heart failure	-Boot shaped heart (X-ray) -Right to left shunt	-Right ventricular hypertrophy -VSD -Over-riding aorta -Pulmonary stenosis
Ventricular septal defect (VSD)		-Older age presentation -Trisomy 21, 13, 18 (ass.) -Asymptomatic initially	-Jet lesions -Right heart failure -Pulmonary HTN -Pulmonary embolism	-Pansystolic murmur loudest at left sternal border -Eisenmengers syndrome -Cyanosis due to shunt reversal -Finger clubbing -Polycythopenia	-Membranous ventricular septum. -Initially L→R shunt but can reverse to R→L shunt.
Atrial septal defect (ASD)		-Older age presentation -Asymptomatic initially -Fix splitting of S2 -Down syndrome (ass.) -Ostium primum	-Pulmonary HTN -Paradoxical emboli	-NO jet lesions d/t ↓ pressures	-Atrial septum: ostium primum, ostium secundum, and sinus venosus. -ASD = Adults
Pericarditis -Serous -Fibrinous -Suppurative		-Uremia (most common) -Post-MI -Pain that's relieved by leaning forward -Asculation-friction rub	-Cardiac tamponade -Constrictive pericarditis	-ST elevation -Dresslers autoimmune	-Pericardium. -Other causes: Coxsackie, TB, metastatic cancer, radiation, SLE, MI, rheumatic.
Rhabdomyoma		-Children -Mutation in TSC1/TSC2 -Tuberous sclerosis ass.	-None	-Spontaneous regression	-Benign hamartoma of cardiac m. **Cardiac tumors are typically due to metastatic neoplasms from lung of breast involving the pericardium.
Cardiac myxoma		-Adult -Left atrium affected -Asymptomatic		-Multinucleated stellate cells -Mucopolysaccharide stroma	-Mesenchymal benign tumor.

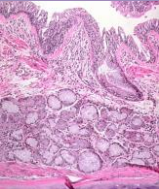
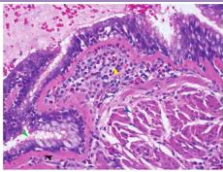
Disease (3)	Slides	Presentation	Complications	Key Words	Area Affected/Other
Pulmonary artery embolism		-Massive: Sudden death -Major: Dyspnea, pleuritic chest pain, V/Q mismatch -Minor: Asymptomatic	-Massive: death -Major: Pulmonary hemorrhage or red infarct in lower lobes -Minor: RHF w/out LHF	-Saddle embolus (massive) -DVT -Spiral CT -Patient outta bed 1st time	-The lungs (Bifurcation of pulmonary arteries, medium vessels, or small). -Embolism commonly from femoral, popliteal, or iliac artery.
Fat embolism		-1-3 days after long bone fracture -Trauma to subcutaneous tissue -Burns	-DIC -Nephrotic syndrome	-Fat globules in urine -Fat globules in sputum -3 DAYS LATER! -Yellow bone marrow	-Endothelial cells of vessels by FFAs.
Bone marrow embolism		-Post aggressive CPR -Flat bone injury -Incidental death	-NONE	-Red bone marrow -Hematopoietic cells -Immediate!	-Small pulmonary vessels.
Amniotic fluid embolism		-Post-delivery -Dyspnea	-Diffuse alveolar damage -Massive DIC -Death	-Fetal squames, hair, meconium -Swirls on histology slide -Rupture of uterine veins	-Pulmonary artery.
Atheromatous emboli		-Asymptomatic -Incidental finding	-NONE	-Cholesterol crystals -Needle-shaped clefts -Foam cells -Intact endothelium	-Peripheral arteries.
Systemic thromboembolism		-Infarct of lower extremity	-Stroke -Gangrene	-Paradoxical emboli -Heart origination	-End arteries of lower extremities and brain.
Nitrogen embolism (The bends) (Caisson's disease)		-Recent deep sea diving >10m -Surfaced too quickly	-Tissue rupture -Brain death -Join "bends" -Pulmonary edema/hemorrhage	-N₂ bubbling out -Deep sea diving -Pressure chamber needed	-Vessel tissues. In Caisson's gas emboli in bones.
Air embolism		-Thoracentesis -Cut jugular vein -Hemodialysis -Child birth/abortion	-Death >150 mL	-Frothy mixture in RV -Air bubbles - H₂O autopsy	-Occlude large vessels or obstruct flow to RV, lungs, and brain.

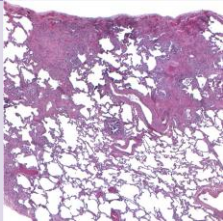
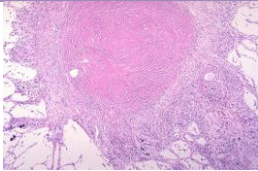
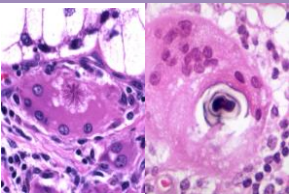
Diseases (4)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Furuncle		-Single infected follicle/gland -Acute inflammation signs	-Carbuncle formation -Fibrotic healing	-S. aureus/S. pyogenes -Liquafactive necrosis (pus) -Streptokinase/hyaluronidase	-Dermis and epidermis -Differentiate from folliculitis because the gland affect too here.
Carbuncle		-Multiple furuncles -At nape of neck -Uncontrolled diabetics -Acute inflammation signs	-Skin infection -Infection spread to other -Abscesses	-Diabetic ketoacidosis -S. aureus/S. pyogenes -Liquafactive necrosis (pus) -Streptokinase/hyalurnidase	-Dermis and epidermis -Communicating furuncles
Impetigo		-Children -Red sores that itch -Honey colored crust on face	-Hematuria (~2 wks) -Post-strep glomerulophritis	-Typical resolution -S. pyogens/S. aureus -Hyaluronidase/collagenase/SK	-Epidermis only (no scarring). -Type III hypersensitivity. -Inoculation.
Bacterial Meningitis		-Headache -Fever -Stiff neck (nucal rigidity) -Photophobia -Papilledema	-Uncal herniation -Encephalitis -Untreated→deafness	-CSF: ↑ Protein/↓ glucose/↑ PMNs -Neo-2mth: M.A.C. -Infancy-Childhood: P.I.M. -Adolescence: N. meningitides -20-60: P.M.M.B(-).	-Inflammation of leptomeninges (pia, arachnoid, and ara space), -Inhalation of respiratory droplets. -Differentiate from other causes of meningitis by CSF and ↑ CD8+!
Meningococcal Septicemia		-Purpuric eruption -Hypotension -High fever -Thrombocytopenia -Buccal/conjunctiva hem.	-Waterhouse Friderichsen -Septic shock -Addison's disease -Hyperkalemia & arrhythmias -Multiple organ failure	-E. coli, H. influenza, P. aeruginos, Klebseilla, Proteus, Serratia. -LPS-LBP in blood→TNFα→IL-1 -↑ bleeding time -Elevated D-dimers	-Gram(-) bacteria endotoxin that causing DIC and eventually septic shock. -TNFα: Endothlium damage -IL-1: Pro-coagulant
Diphtheria		-Unvaccinated children -Pseudomembrane -Sore throat -Fever w/ malaise -Bull neck (common COD)	-Suffocation -Respiratory distress -Heart failure (3wks) -Peripheral neuropathy (3wks) -Exotoxic shock (scrapping)	-Corynebacterium diphtheria (+) -ADP-ribosylation of EF2 -Necrosis of oropharyngeal cells -AB exotoxin (causes symptoms)	-Severe throat infection that can be prevented by DTaP vaccination. -Inhalation of respiratory droplets containing bacteriophage.
Actinomycosis		-Dental trauma -Cervico-facial -Fever -Painful lumpy jaw -Yellow sulfur granules -Thoracic -Pneumonia	-Fibrosis -Abscess (common in alcoholics)	-Actinomyces israelii (commensal) -Filamentous branching gram (+)	-Infection of the cervico-facial, thoracic area, abdominal area (direct inoculation from surgery), or pelvic area (contaminated intrauterine devices). -**Sulfur granules DON'T contain sulfur!!**
Cytomegalovirus (HHV-5)		-Immunocompromised -CMV retinitis -Interstitial pneumonia -Esophagitis/gastritis/colitis -Transplant: nephritis/hepatitis	-Pneumonitis (most common)	-Cotton wool app. of retina -Intranuclear basophilic inclusions ^Owl eye inclusions -Intracytoplasmic inclusions -Marginated chromatin	-Contraction from body fluids or vertically that is initially asymptomatic but polyclonal B-cell activation that presents when immunocompromised.
Herpes Zoster (Chicken pox) (Varicella Zoster)		-Unilateral, painful vesicles -Follow single dermatome -Stressed pt -Immunocompromised	-Post-herpetic neuralgia -Vision loss -Hearing loss -Vertigo	-Tzanck smear w/ giemsa -Multinucleated (syncytia) cells -Intranuclear eosinophil inclusion ^Cowdry type A	-Dormant in DRG till ↓ immunity. -Inhalation of respiratory droplets. *3M's: <u>M</u>argination, <u>M</u>olding, and <u>M</u>ultinucleated.

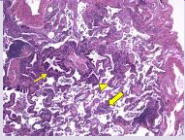
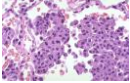
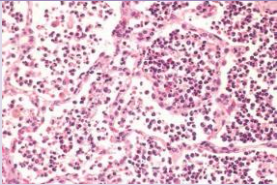
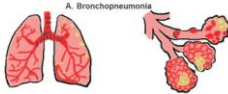
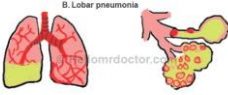
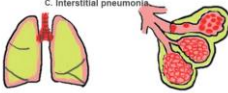
Diseases (4)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Rabies <i>Differentiate from Herpes zoster due to intracytoplasmic eosinophilic inclusions</i>		-Animal bite (bats/dogs) -Early: flu-like symptoms -Loss of consciousness -Hydrophobia -Spasms of pharyngeal m.	-Respiratory failure -Coma -Death	-Negri bodies in neurons -Motor neurons affected	-Virus spreads to PNS/CNS by retrograde axonal transport. -Prognosis based on bite location. -Inoculation by bite or acquired by inhalation.
Rocky Mountain Spotted Fever		-Camper/hiker -Recent tick bite -GOAT regions -Fever w/ myalgia -Headache -Centripetal rash	-Gangrene -Stroke -Red infarct of lung or brain -Coagulative necrosis	-Rickettsia rickettsia -(+) Weil-felix test -Immunofluorescent skin stain -Mφ first (NOT neutrophils)	-Virus with predilection for endothelial cells and smooth m. of arterioles. -Differentiate: -R. prowazeki has centrifugal rash. -C. burnetti has (-) Weil-felix test.
Candida albicans		-Opportunistic -Asthmatics (corticosteroid) -Diabetic patients -Immunocompromised -Broad spectrum Ab use -Dimorphic fungi	-Invasive form – abscesses -Candida endocarditis	-Budding yeast w/ pseudohyphae -Chinese letter appearance -Stains: -PAS, Silver Stain, Mucicarmine -Vaginal: cottage cheese app. -Low temp: yeast, High temp: mold	-Can occur in mouth, esophagus, or vagina and is usually superficial.
Pneumocystis Jiroveci Pneumonia		-Opportunistic (CD4 <200) -Fever -Dyspnea on exertion -Dry cough -Interstitial pneumonia		-Cotton candy exudate (acellular) -Black cup shaped cyst -Bronchoalveolar lavage	-Inflammation and thickening of alveolar wall. -Inhalation of spores. **Anti-fungal don't work.
Aspergillus		-Immunocompromised -Asthma exacerbation -Septated hyphae w/ acute	-Aspergilloma (immunocompetent) -Allergic bronchopulmonary aspergillosis -Dissemination aspergillosis -Necrotizing pneumonia -Stroke -Coagulative necrosis -Hepatocellular cancer (aflatoxin)	-A. flavus (aflatoxin) -A. fumigatus (gliotoxin) -Curschman spiral (asthma) -Charcot-Leyden crystals (asthma) -Stain: -PAS, silver, mucicarmine -Giant cells present	-Inhalation of mycelial form aspergillus affects the lungs and brain, they are <u>angio-invasive</u> . -Allergic bronchopulmonary aspergillosis. ^Type I and III hypersensitivity.
Mucormycosis		-Immunocompromised -Diabetic ketoacidotic pts -Sero-sangrenous nasal dis. -Fever & Pain -Headache -Rhino-facial-palatal necrosis	-Red infarct of sinuses, brain, GI, lung.	-Mucor, Rhizopus, & Absidia -Broad organisms w/ non-septate hyphae branching @ right angles. -Ribbon like app.	-Inhalation of spores travel to endothelial cells of blood vessels in cerebral hemisphere, lung, GI, sinuses where its <u>angio-invasive</u> . -Differentiate from aspergillus due to non-septate, 90° branching .

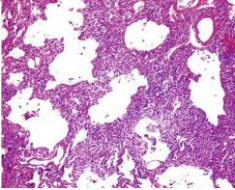
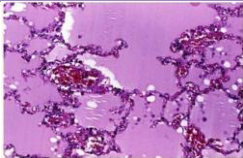
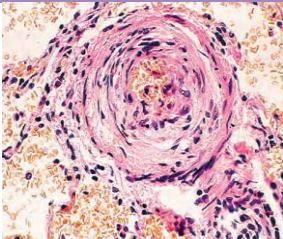
Diseases (4)	Slides	Presentation	Complication	Key Words	Area Affected/Other
Cryptococcus Neoformas (Torulosis)	 	-Immunocompromised -HIV: (CD4 <100) -Pigeon dropping contact -↑ICP: photophobia, headache projectile vomiting	-Meningitis if disseminates	-Soap bubble lesions in brain -Encapsulated fungi -Stains: India ink (clear halo) -PAS, silver stain -Mucicarmine (red, 2 nd pic) -Granulomas with giant cells	-Inhalation of spores from pigeon droppings→lungs and brain affected. -Unlike bacterial meningitis shows NO fever or neck stiffness.
Tuberculosis (1°, 1° progressive, 2°, 2° progressive, miliary) <i>PPD >15: no exposure</i> <i>PPD >10: risk present</i> <i>PPD >5: HIV/AIDS, ect.</i>		-HIV/immunocompromised -Recent Asia or Africa trip -Fever -Night sweats -Weight loss -Cough w/ hemoptysis	-Miliary TB→extensive necrosis -Encephalitis -Pancytopenia -Infertility (testes or ovaries) -Renal failure -Lung abscess→empyema -Late: AA amyloidosis -Pott's disease	-Milky, cheese app. -Gohns complex -Acid fast stain (mycolic acid) -1°: Very mild/contained (Gohn) -2°: Get symptoms (apex) -Langhan's giant cells (exclusive)	-Inhalation of M. tuberculosis→caseating granuloma -Type IV hypersensitivity. -Mφ first (IL-12)→TH1 (INFγ)→Mφ to <u>Epitheloid cells</u> (TNF & IL-1)
Histoplasma Capsulatum	  	-Acute: Immunocompetent -Caseating granuloma (Up) -Chronic: Immunocompromised -Cough w/ hemoptysis -Fever -Night sweat -Weight loss -OH, Appalachia, MS river valley -Recent cave hiking	-Acute: Complete resolution -Chronic: Mimics 1° TB -Hepatosplenomegaly -Lymphadenopathy -Pancytopenia	-Buckshot dystrophic calcification -↑ Mφ pH (allows yeast budding) -Mφ w/ <u>intracytoplasmic</u> budding	-Inhalation of spore in bird/bat droppings. -Differentiate from Blastomyces (includes great lakes) because Blasto is a broad based budding. -Differentiate from TB because in competent see buckshot calcification, unlike TB. And see PMNs. -Differentiate from TB because in compromised see dissemination goes to RES, unlike TB.
Cat Scratch Disease		-Immunocompetent: -Fever w/ malaise -Headache -Tender, warm cervical & axillary lymph nodes	-Immunocompromised: -Osteomyelitis -Encephalitis -Thrombocytopenia -Death	-Bartonella henselae, gram (-) rod -Suppurative granuloma (PMNs) -Stellate area of necrosis	-Non-caseating, necrotizing granuloma from inoculation from a bite or scratch from a cat/kitten. *Caseating means looks cheese-like to unaided eye -- Wikipedia
Tubercloid Leprosy <i>Differentiate tuberculoid from lepromatous because tub. Has (+) test and asymmetric PNS involvement.</i>		-Skin lesions (L>T) -Peripheral neuropathy -T: Asymmetrical -L: Symmetrical -Infertility in men -Auto-amputation of digits	-Sepsis due to 2° infection -Death	-Tuberculoid (Good CD4) -Caseating granuloma -Lepromatous (Poor CD4) -Leonine facies -Virchow/foamy/lepra cells -Acid fast rods in Mφ	-Inhalation of M. Leprae that can cause tuberculoid (cellular) or lepromatous (IL-4 & IL-10; humoral) reaction. -In Lepromatous theres no IFNγ so no granuloma formation. -T: Type IV hypersensitivity.

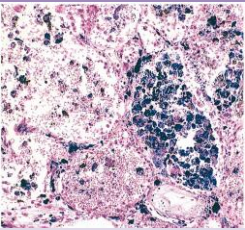
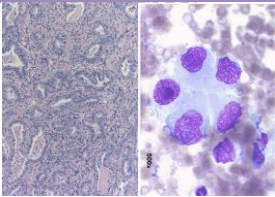
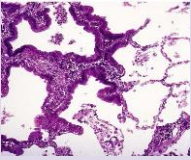
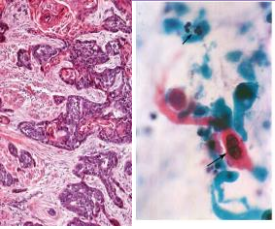
Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
Atelectasis (resorption)		-Shortness of breath -Ipsilateral tracheal deviation on CXR -Decreased breath sounds -Dullness on percussion	-Hypoxemia -Pneumonia -Respiratory failure -Death	-Reversible -Mucus plug -Intrinsic obstruction	-Bronchial lumen is obstructed. -Air in blocked area gets absorbed causing collapse.
Atelectasis (compression)		-Shortness of breath -Contralateral tracheal deviation on CXR -Decreased breath sounds -Dullness on percussion	-Hypoxemia -Pneumonia -Respiratory failure -Death	-Anything in pleural cavity (air, fluid, ect.) -Reversible -Extrinsic obstruction	-Compression of adjacent lung due to increased pressure in the pleural space.
Atelactasis (contraction)		-Shortness of breath -Decreased breath sounds -Dullness on percussion -Have a restrictive lung disease -Ipsilateral tracheal deviation on CXR -Spirometer: ↓ TLC & RV	-Hypoemia -Pneumonia -Respiratory failure -Death	-Irreversible -Pulmonary fibrosis -Normal FEV1:FVC ratio -Inc. elastic recoil	-Lung parenchyma or pleura undergoes fibrotic changes -Fibrosis can be local or generalized.
Atelactasis (neonatal)		-Premature infant <34 weeks -Cyanotic -Nasal flaring	-Respiratory distress syndrome -Hypoemia -Respiratory failure -Death	-Mom is a diabetic -C section delivery	-Alveolar collapse do to increased surface tension caused by lack of surfactant. -Give exogenous steroids to mother to help type II pneumocytes to mature.
Acute Respiratory Distress Syndrome (ARDS) [Active/Exudate]		-Blunt force trauma to chest -Almost drowned -Acute pancreatitis (indirect) -<u>Sudden onset</u> -Tachypnea (SOB) -Tachycardia -Cyansosis (doesn't improve with O₂) -<u>Normal wedge pressure</u>	-Respiratory failure -Fibrosis→con. - Atectasis -<u>Pulmonary edema</u>→RHF -Cor pulmonale -Chronic passive congestion -Nutmeg liver due to RHF -Mechanical ventilation	-Ground glass app. on x-ray -Exudative stage 0-7 days: -Neutrophils, debris, fibrin -Hyaline membrane -Not seen in neonatal RDS -Proliferative stage 1-3 weeks: -↑ # of type II and fibroblasts -alveolar septal thickening	-Damage to type I pneumocytes and endothelial cells. -Imbalance of pro- and anti-inflammatory responses. -4 criteria must be meet: sudden onset, ground glass app., pulmonary edema w/ normal wedge pressure, and <u>refractory hypoxemia</u>.

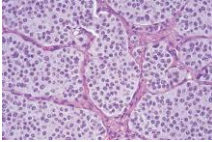
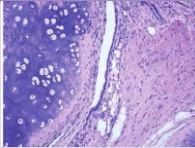
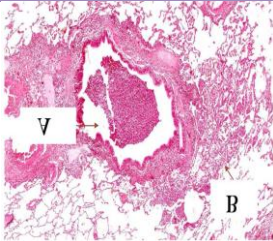
Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
Emphysema		-Smoker #1 (centroacinar) -A1AT deficiency (panacinar) -Child with liver cirrhosis -Skinny with barrel chest -Pursed lip breathing -↓FEV1:FVC ; ↑TLC & RV	-Respiratory failure -Cor pulmonale (RHF) -Secondary polycythemia -Liver congestion/cirrhosis -Compression atelectasis -Septal (distal acinar)	-Functional obstruction -Pink puffers (2° polycythemia) -CXR: (hyper-lucent) -Hyper-inflated lungs -Flattened diaphragm -Narrow mediastinum	-Airways distal to terminal bronchioles will undergo destruction/dilation w/out fibrosis. -Irreversible obstructive disease. -Exception: irregular (paracicratrial) has fibrosis→contract. atele.
Chronic Bronchitis		-Productive cough -Foul smell/purulent sputum -3 consecutive mths/2 con. yrs -Smoker -↓FEV1:FVC ; ↑TLC & RV	-Bronchiolitis obliterans -Ascites/peripheral edema -Cor pulmonale -Respiratory failure -Passive liver congestion -Resorption atelectasis	-Mechanical obstruction -Reid index >50% -Blue bloaters	-Affects <u>bronchus</u> (goblet cell hyperplasia) and <u>bronchioles</u> (goblet cell metaplasia). -Squamous metaplasia also present (unlike atopic bronchial asthma where is absent!).
Atopic Bronchial Asthma		-Episodic (family history) -Expiratory wheeze -Chest tightness -Productive cough -Shortness of breath -↓FEV1:FVC ; ↑TLC & RV	-Respiratory failure -Mucus plug formation -Resorption atelectasis -Status asthmaticus -Bronchiectasis	-Curschmann spirals -Charcot Layden crystals -Eosinophilia (IgE/IL-5) -Reid index >50%	-Affects the bronchi causing bronchoconstriction. *ONLY reversible obstructive airway disease (clinical presentation). -Type I hypersensitivity. -No squamous metaplasia.
Non-atopic Bronchial Asthma		-Aspirin use (intrinsic) -Nasal polyps -Intrinsic asthma -Allergic rhinitis -Bronchoconstriction -No family history	-Status asthmaticus -Doesn't respond to med -Intubation	-No eosinophils or IgE -No charcot leyden crystals -Curschman spirals/neutrophil -Viral: ↑PS response -Aspirin: ↑ leukotrienes	-Bronchi causing constriction that can be treated with albuterol (β_2 agonist).
Bronchiectasis		-Cystic fibrosis -Kartageners syndrome -Chronic bronchitis, ect. -Halitosis -Prod. Cough w/ foul sputum -Finger clubbing (hypoxia) -Hemoptysis	-Respiratory failure -Cor pulmonale -Metastatic abscess/sepsis -AA amyloidosis -Restrictive cardiomyopathy -Resorption atelectasis	-S. aureus or Klebsiella initially -CXR/CT: train track app. -Mixed flora in sputum **Cystic fibrosis**	-Permanent dilation of the bronchi to pleural surface. -2° to an infection.

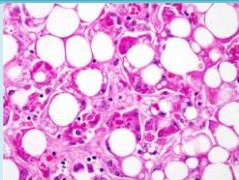
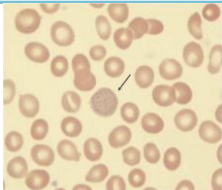
Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
Idiopathic Pulmonary Fibrosis (UIP)		-Male >60 -Smoker -Dry cough -Exertional dyspnea -Finger clubbing -Bilateral fine insp crackles -Norm FEV1:FVC ; ↓TLC & RV	-Lung transplant -Respiratory failure -Contraction atelectasis -Cor pulmonale -Cardiac cirrhosis	-Progressive and relentless -Lower lobe first -Honeycomb lung app. -Cobblestone pleural surface -CXR: bilateral reticulo-nodular shadows -Temporal heterogeneity	-Damages the type 1 pneumocytes which leads to fibrosis.
Non-Specific interstitial pneumonia		-Younger individuals -Cellular: ↑ lymphocytes -Fibrosis: ↑ fibrosis		-Chicken wire fibrosis -Doesn't do honeycombing	-Lung interstitial septa.
Cryptogenic organizing pneumonia	Also called bronchiolitis obliterans organizing pneumonia	-CXR: patchy peribronchial and sub-pleural consolidation		-Masson bodies -Spontaneous recovery	-Alveolar ducts, alveoli, and bronchioles are plugged with loose CT but architecture NOT destroyed!
Lymphoid interstitial pneumonia		-Present CT disease -Autoimmune disease -HIV	-Lymphoma	-Expansion of interstitium	-Interstitium of lung affected.
Coal workers pneumoconiosis		-Individuals in urban area -Smoker -Coal worker -Exertional dyspnea -Non-productive cough -Norm FEV1:FVC ; ↓TLC & RV	-Contraction atelectasis -Cor pulmonale -Focal dust emphysema	-Black lung disease -Pulmonary anthracosis -Honeycomb lung (if impure)	-Black pigment accumulation in the lungs macrophages.
Silicosis		-Miner/quarry worker -Glass manufacturer -Stone cutter -Exertional dyspnea -Non-productive cough -Norm FEV1:FVC ; ↓TLC & RV	-Contraction atelectasis -Respiratory failure -Cor pulmonale -Pulmonary TB	-Upper lobes first -Whorls of collagen nodules -Eggshell calcification on hilar -White birefringent particles	-Crystalline silica engulfed macrophages that die due to damaged cell membrane.
Asbestosis		-Miner/mill worker -Shipyard/ship workers -Construction/insulation -Asbestos bodies in sputum -Dry cough -Exertional dyspnea	-Contraction atelectasis -Cor pulmonale -Bronchogenic cancer -Mesothelioma (unique) -Pleural plaques	-Honeycomb lung -Lower lung -Mesothelioma -Ferruginous bodies -Beaded structure	-Diffuses interstitial fibrosis around bronchioles, alveolar ducts, and interstitium. -Mesothelioma: rind-like encasement of lung, stain for calretinin, 3 patterns (epithelial, sarcomatoid, biphasic).
Sarcoidosis		-African Am., Danish, Swedish -Non-smoker -Young female -Dry cough -Exertional dyspnea -Hypercalcemia & ↑ ACE	-Restrict. cardiomyopathy -Sicca syndrome -Milchlit syndrome -Pulmonary failure -Cor pulmonale -Contraction atelectasis	-Non-caseating granuloma -Honeycomb lung -Schaumann body -Asteroid bodies -Metastatic calcification -Elevated ACE	-Starts as non-caseating granuloma in lungs then becomes multi-system. -Type IV hypersensitivity. -*Vitamin D activated by epithelial cells → hypercalcemia!

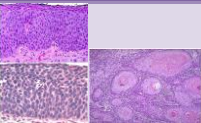
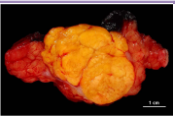
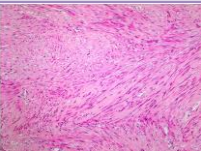
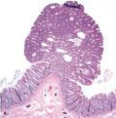
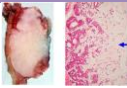
Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
Hypersensitivity Pneumonia (Extrinsic allergic alveolitis)		-Non-necrotizing granuloma -Farmers -Pigeon breeder -Air humidifier/AC -Giant cells	-Restrictive lung disease	-Extrinsic organic antigen -Airway centered process	-Damage at level of alveoli. -Type III and IV hypersensitivity.
Desquamate Interstitial Pneumonia		-Smokers -Minimal interstitial fibrosis		-Macrophages fill alveolar spaces	-Alveolar spaces. -RARE.
Respiratory Bronchiolitis		-Smokers -40-50 yo -Associated with perbronchiolar fibrosis		-Smoker's macrophage: carbon accumulation (black)	-Respiratory bronchioles get accumulation of smoker's Mφ. -Responds to steroid/quitting.
Pulmonary Langerhan Cell Histiocytes	Also called Eosinophilic Granuloma	-SMOKERS ^Almost exclusively! -Langerhans cells (culprit)		-Birbeck granules on EM -Tennis racket appearance	
Typical Lobar Pneumonia		-Young adults -Immunocompetent -S. pneumoniae infection -Fever w/ malaise -Pleuritic chest pain -Productive cough (bloody) -CXR: lobe consolidation	-NONE -Complete resolution -10-14 days later	-S. pneumoniae/Klebsiella -Stages: -Congestion -Red hepatization -Gray hepatization -Resolution -Intact architecture	-Consolidation of entire lobe due to spreading through pores of Kohn involving alveolar spaces (no damage). -No bronchial involvement. -Lower lobe or right middle lobe.
Bronchopneumonia (Typical)	  	-Klebsiella -Alcoholics -Diabetics -Pseudomonas -Cystic fibrosis -Ventilators -Burns -Lung abscesses -S. Aureus -Following viral infection -Legionella -AC at hotel -Confusion -Diarrhea/vomiting -Abdominal pain -Liver/Kidney issues	-Lung abscess -Aspergilloma -Empyema -Compression atelectasis -Recurrent pneumonia -Scarring	-Patchy lung -Klebsiella -Red currant jelly sputum -Angio-invasive pneumonia -Pseudomonas -Angio-invasive pneumonia -Legionella -(+) urine antigen	-Damage to bronchi and bronchioles.
Community Acquired Acute Pneumonia	-S. pneumonia -H. influenzae-#1 in COPD -Moraxella catarrhalis -S. aureus -K. pneumoniae -P. aeruginosa -L. pneumophila	-Age extreme (young/old) -Immunocompromised -Asplenic -High grade fever -Shaking chills -Pleuritic chest pain -Mucopurulent sputum	-S. Aureus: acute endocarditis	*Each has their own!	-Bacterial causing local OR focal consolidation of the lung.

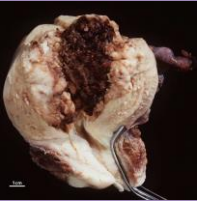
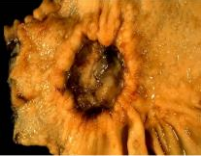
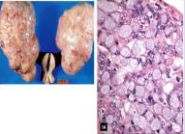
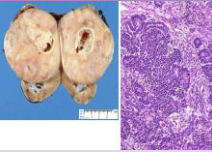
Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
Community Acquired Atypical Pneumonia <i>Most common: (Mycoplasma)</i>		-Children and young adults -Dry cough -Low grade fever -Hyaline membrane (DAD) -CXR: interstitial infiltrates -No dullness on percussion	-ARDS -Autoimmune hemolytic anemia	-Walking pneumonia -CXR: worse than prensenting -Ground glass app. -CXR: reticular opacities -Lymphocytes/plasma cells -Coxiella: Ring granuloma -Myco: Cold agglutination (+)	-Inflammation of alveolar wall causing thickened septae. -Air spaces clear. -Also viral cause: CMV, HPIV. -Other bacteria: Chlamydia and Coxiella burnetti.
Lung Abscess <i>Most common: (S. Aureus)</i>		-Alcoholics -Stoke patients (↓ gag reflex) -Intubated/hosp. bed ridden -Seizure -Alzheimer's -IV drug user (septic emboli) -Prod. cough-hemoptysis -High spiking fever -Halitosis	-Empyema (d/t rupture) -Compression atelectasis -Bronchopleural effusion -Septicemia -AA amyloidosis ^If have nephrotic syn. -Aspergilloma -Anemia	-Sputum: mixed flora -FINGER CLUBBING -CXR: Air fluid level	-Aspirate bacteria more common on right side. -Standing: bottom of r. lung. -Laying: top of r lung. -Liquefactive necrosis. -Differentiate from bronchiectasis due to >6 weeks and spiking fever!
Cardiogenic Pulmonary Edema		-Dyspnea/orthopnea -Paroxysmal dyspnea -Frothy sputum(>2wks pink) -Hypoalbuminemia -↑ wedge pressure (LHF) -Left ventricular hypertrophy	-Pulmonary HTN -Right heart failure -Nutmeg liver -Cardiac cirrhosis	-Hemosiderin laden Mφ -Aka heart failure cells	-Fluid in alveolar spaces causing decreased diffusion. - <2 wks: Transudate - >2 wks: Fibrosis/HF cells
Pulmonary Embolism		-History of DVT/birth control -Recent long flight -Factor V leiden mutation -Adenocarcinoma (procoag.) -Dyspnea & hemoptysis -Pleuritic chest pain	-Death -Coagulative red necrosis -Cor pulmonale -Nutmeg liver	-Virchows triad: -Hypercoagulability -Turbulence -Endothelial injury -Compromise of bronchial circulation = Major w/ CHF	-Can black anywhere from bifurcation to smaller vessels: -Saddle (massive)-sudden death. -Major w/ CHF-Red infarct. -Major-pulmonary hemorrhage. -Minor-asymptomatic
Pulmonary Hypertension <i>Avg pul > 1/4 systemic</i>		-1° BMPR2 mutation -Young women 20-40 -Plexiform lesions -2° -Resiratory insufficiency d/t underlying disease -Syncope, angina, dyspnea -S2 spilt w/ loud P2	-Right heart failure -Nutmeg liver -Atheroma d/t HTN ^Pulmonary trunk -Fibrinoid necrosis (S.V.) -Pitting edema/ascites -Death	-Reduplication of IEL & EEL in small vessels.	-Increased blood pressure leads to changes in lungs and blood vessels. -Bronchial asthma does NOT cause!

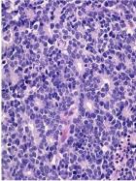
Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
Diffuse Pulmonary Syndrome (Goodpastures) <i>Idiopathic pulmonary hemosiderosis is similar but <u>NO</u> renal involvement</i> -Children>adults -Self-limiting		-↑ wedge pressure -Young men -Hematuria (g. nephritis) -Patchy necrosis of walls -Septal thickening -Triad: -Iron deficiency anemia -Cough w/ hemoptysis -Pulmonary infiltrates	-Death if untreated	-Hemosiderin laden Mφ -Linear deposits of IgG & C3 -Temporal heterogeneity -Anti-glomerular-BM Ab	-Active bleeding into lungs w/ fibrotic thickening of alveolar septa – Differentiate from Wegner's because Goodpasture's is NEVER seen in the URT! Also, use age! -Type II hypersensitivity. ^Unlike Wegener's type IV.
Adenocarcinoma <i>Has a para-neoplastic syndrome (Hypercoagulable)</i>	 Acinar gland formation	-Non-smoker -Female <45 yo -Asian -Dyspnea -Cachexia (TNFα) -Finger clubbing -CXR: peripheral coin lesion -Trousseau syndrome	-Pleural effusions -Compression atelectasis -Pericardial effusion -Tamponade	-Early metastasis -Scar carcinoma -Stains: -TTF-1 -cytokeratin 7 -napsin-A -PAS (+) d/t Mucin -Trousseau-hypercoagulable	-Bronchiolalveolar stem cell/clara cells undergo atypical adenomatous hyperplasia. -Lung→region lymph nodes→adrenal glands→liver→brain→bone→kidney. -EGFR: Asian women <45, nonsmoker -ALK: Signet, non-smokers. -KRAS: Smokers.
Adenocarcinoma in-situ		-Productive cough (sputum) -Shortness of breath -Pleuritic chest pain -CXR: lobar consolidation	-Respiratory failure -Asphyxiation on mucus	-Lepidic pattern on biopsy -Butterfly on fence app. -Ground glass app.	-Overproduction of mucins spread through pores of Kohn. -Differentiate from adenocarcinoma because its non-invasive, no coin lesion, and no para-neoplastic syn.
Squamous Cell Carcinoma Bronchogenic carcinoma / bronchial cancer <i>Has a para-neoplastic syndrome (PTH-releasing peptide)</i>		-Smoking + asbestosis -Men > women -Raydon, p53, Rb mutation -Dyspnea w/ hemoptysis -Cachexia -Finger clubbing -Hoarseness of voice -Hypercalcemia (PTH-like)	-S: SVC syndrome (red face) -P: Pancoast tumor -H: Horner's syndrome -E: Endocrine paraneoplastic -R: Recurrent laryngeal -E: Effusions (either) -Bronchiectasis -Abscess -Resorption atelectasis	-Keratin pearls -Desmonin bridges (inter-) -Ind. cell keratinization -Stains: P63 -Cytokeratin 5/6 or 903 -Tadpole cells (dysplastic) -Widening of mediastinum	-Pseudostratified→squamous metaplasia followed by squamous dysplasia→ central carcinoma. -Unlike adenocarcinoma its centrally located w/ hypercalcemia and keratin tadpoles. -4C: <u>C</u> entral, <u>C</u> avities, <u>C</u> igarettes, hyper <u>C</u> alcemia.
Small (oat) Cell Carcinoma <i>Has a para-neoplastic syndrome (SIADH-hyponatremia) (Gonadotropins)</i>	 Gynomastia d/t gonadotropins Purple Striae d/t cushing syn.	-Smoking (almost always) -SIADH: -Cerebral edema -Papilledema -Gonadotrophins -LH-amenorrhea -Cushing syndrome -Buffalo hump -Moon face	-Metastasis -Death	-Salt & pepper app. (biopsy) -Nuclear molding -Rim/dot staining -Neuroendocrine markers: -Synaptophysin -Chromogranin -CD56	-Kulchitsky cells cause a centrally located carcinoma. -Differentiate for bronchial carcinoid because small cell shows molding. Based on Cushing can differentiate from pituitary causing due to ACTH levels in plasma. *Liver and kidney show NO sym.

Disease (Res)	Slides	Presentation	Complications	Key words	Areas affected/other
(Cushing syndrome)					
Bronchial Carcinoid Has a para-neoplastic syndrome (Carcinoid - ↑ serotonin)		-Episodic flushing -Wheezing -Diarrhea	-Resorption atelectasis	-5-HIAA in urine -Neuroendocrine markers: -Synaptophysin -Chromogranin -CD56	-Kulchitsky cells lead to centrally located, well-nested carcinoma that looks histologically the same as small cell. *Metastasis is RARE.
Pulmonary Hamartoma		-Asymptomatic -Incidental finding		-Popcorn calcification -CXR: coin lesion	-Normal tissue, disorganized growth forming a benign nodular neoplasm that can be centrally or peripherally located. -Made of CT and cartilage mostly.
Lobular broncho-pneumonia		-Children and elderly -Bed-ridden -Immunocompromised -Hospitalized pts -Fever -Dyspnea w/ hemoptysis -Pleuritic chest pain	-Sepsis -DIC -Lung abscess -Aspergilloma -Empyema -Pleural fibrosis -Scarring	-CXR: patchy consolidation -Circular patches -Klebsiella, E.coli, Staph, Pseudomonas	-Starts in terminal bronchus and spreads to alveoli, destroying bronchiolar walls. -Commonly follows an URT infection. -Differentiate from lobar b/c there are NO Mφ in the bronchi (scarring), where lobar has Mφ in alveoli.

Environmental	Slide	Presentation	Complication	Key words	Areas affected/other
Alcoholic Hepatitis (liver steatosis)		-Recent alcohol consumption -AST>ALT (scotch over tonic) -Jaundice -Fever -Vomiting -Right upper quadrant pain	-Cirrhosis (irreversible) -Hepatocellular carcinoma	-Mallory bodies (ubiquitinated keratin intermediate filaments) -Reversible if cease alcohol use -Intracellular lipid accumulates -Fatty liver -Nucleus (distinguish from CCl₄)	-Liver for alcoholic (but fatty change can occur in heart and kidney as well). -↓ NAD⁺ due to neutralization of ethanol leading to fat accumulation because NAD⁺ is needed for β-oxidation of fatty acids in liver.
Lead Poisoning		-Children more susceptible -Pallor/tachycardia -Lead lines on X-ray -Hyperpigmentation of gums -Children: CNS (Retardation) -Adult: PNS (foot/wrist drop) -Excruciating abdominal pain	-Fanconi syndrome -Anemia	-Basophilic stripping -Microcytic hypochromic w/ mild hemolytic anemia -Blue gums	-Main accumulation in bone/teeth but affects the RBCs, bone, CNS/PNS, kidneys, and GI. -Differentiate from PAN, which also causes foot drop, by measuring blood lead levels.
Carbon Monoxide Poisoning		-Flu-like symptoms -Cherry red discoloration	-Globus Pallidus necrosis -Parkinsonian symptoms -MI -Coma/death	-NORMAL O₂ saturation -Carboxyhemoglobin -Hyperbaric chamber treatment	-Brain affected first (basal ganglia and lenticular nuclei) then affects the heart.
Cocaine Use <i>Sympathomimetic</i>		-Perforated nasal septum -↑ ICP- -Papilledema/vomiting -Retinal hemorrhage/detach -Dilated pupils -Recent cocaine use	-Coagulative necrosis -Dilated cardiomyopathy -Malignant HTN -Aortic dissection -MI or LHF	-↑ Dopamine→movement -↑ NE→Peripheral constriction	-Can affect the CNS/PNS, eyes, aorta, heart, kidney, and lead to secondary infection.
Reye Syndrome (Fatal childhood hepatoencephalopathy)		-Aspirin treatment of: -Chicken pox -Viral infection (flu) -Seizures/confusion -Hypoglycemia -Hepatomegaly	-Cytotoxic cerebral edema -Microvesicular fat change -Coma -Death	-Elevated blood ammonia -Elevated liver enzymes -Urea cycle disruption	-Increased ammonia that starts affecting the liver→brain. -NO acute inflammation! -Differentiate hepatic fatty change from alcohol hepatitis because the nucleus will be centrally located.

Neoplasms	Slides	Presentation	Complication	Key words	Areas affected/other
Cervical Dysplasia to Carcinoma		-HPV 16 & 18 (+ to genome) -Ab. Pap smear (screening) -Multiple sex partners -Post-coital bleeding -Dyspyruria -Leukorrhea	-Chronic renal failure (metastasis)	-Increased nucleus:cytoplasmic -Nuclear hyperchromasia -Nuclear pleomorphism -Cellular pleomorphism -Enlargement/irregularity/loss of polarity -Keratin pearls present	-Cervix. -HPV 16 & 18 DON'T cause metaplasia but go straight to dysplasia. Metaplasia caused by ↑ in lactic acid production. -E6→p53, E7→Rb
Lipoma		-Adults 40-60 -Painless, mobile lump -Commonly on back -Family history	-None -No malignant transformation	-True capsule -Slow growing benign tumor -No necrosis	-Adipose tissue (mesenchymal origin). -Normal fat cell on biopsy.
Hemangioma		-Children < 3 yo -Red, raised lesion -Buttock or face of newborn -Grow with age then regress	-Bleeding occasionally -Spontaneous regression 7-10 yo -No malignant transformation	-Strawberry lesion -Disorganized blood vessels -Port wine stains (larger lesions)	-Type of hamartoma of disorganized capillary tissue in normal location. -Don't biopsy!!
Leiomyoma (Uterine fibroids)		-Nulliparous African American women -Menorrhagia -Urinary urgency/freq. -Constipation	-No malignant transformation -Iron deficiency anemia -Left heart failure -Coagulative necrosis (red infarct) ^Outgrown blood supply	-Pseudocapsule -Whorling bundles of smooth m. -Well differentiated	-Benign tumors of the myometrium that's growth is dependent on estrogen levels (but estrogen doesn't cause!).
Familial Adenomatous Polyposis (FAP)		-Family history (APC) -Blood in feces -Obstruction: constipation -Commonly asymptomatic	-Mutations in KRAS & p53 lead to adenocarcinoma→100% chance -Left heart failure (Iron def anemia)	-Numerous polyps on left side -Sigmoidoscopy screening at 12 -*FAP shorter word so is left!	-Pedunculated outgrowth of mucosal tissue causing many polyps on the left colon. -Tubular adenoma.
Hereditary Non-Polyposis Colon Cancer (HNPCC)		-MLH1, MSH2, MSH6, PMS2 -Family history of breast, right sided colon, endometrium or ovary ca.		-Cancer affects right side -Microsatellite instability -Colonoscopy screening at 25 -Lynch syndrome	-Few, flat cancer growths with no stalk on right side of colon. -Villous adenocarcinoma.
Colon cancer (Sporadic)		-High fat diet w/ low fiber -Rich in refined carbs -Low in vit. A, C, & E -UC & Crohn's -Left: Bright red blood -Mucus -Left lower quad pain -Right: initially none -Pallor -Elderly male	-Left: -Intestinal obstruction -Right: -Occult bleeding -Iron def. anemia -Spread via lymph→liver→lung	-Tumor marker: CEA ^Follow up NOT diagnosis! -Stain used: cytokeratin -Left side: napkin ring constriction -Right side: exophytic growth	
Benign Mixed Tumor of Parotid Gland		-Painless, lump in jaw -Mobile mass -Dry mouth	-Bell's palsy -Malignant transformation (small) -Recurrence	-Divergent differentiation -Fibromyxoid stroma (mesen.) -Serous acini (epithelial cells)	-Benign tumor from 1 germ cell. -A malignant mixed tumor is a Wilm's tumor.

Neoplasms	Slides	Presentation	Complication	Key words	Areas affected/other
Warthin Tumor		-Male smoker		-Epithelial and lymphoid components	-2 nd most common parotid tumor
Osteosarcoma		-Both eyes removed at young age -Multiple fractures -Children 12-20 -Painful swelling near knee	-Metastasize to lung -Cough w/ hemoptysis -Metastasize to liver also -Invasion of muscle	-X-ray: Codman's triangle shadow -X-ray: Sun burst app. - Mutation of Rb gene -Lacy osteoid	- Malignant tumor of osteoblasts. -Reduces E-cadherin allowing local invasion→type IV collagenase invades ECM→attaches to fibronectin receptor→autocrine motility factor.
Endometrial Adenocarcinoma		-Post-menopausal women -Estrogen replacement -Vaginal bleeding -Leukorrhea -Obesity -Diabetes mellitis -Nulliparous	-Seeding in peritoneal cavity -Invasion of myometrium -Hematogenous spread	-Stain: cytokeratin -PTEN mismatch repair mutation ^Cowden's syndrome -Endophytic (friable) growth	-Hormone producing cancer of endometrium. - Differentiate from fibroids because estrogen is direct cause. -Can occur where-ever glandular tissue is.
Gastric Adenocarcinoma Chronic gastritis causes pro-inflammatory release of IL-1 and TNF α		-Japanese -Smoked food (nitrosamines) -Epigastric pain -Hematemesis -Melena (dark, tarry stools) -Cachexia and fatigue	-Fibrosis -Hemorrhage	-Enlarged Virchow's node -Sister mary joseph nodule -Rugal fold disappearance -Tumor marker: CEA -Irregular, headed up margins	-Chronic atrophy gastritis that leads to adenocarcinoma, associated with H. pylori and B-cell lymphoma. - CEA for CLuBS cancers.
Bilateral Ovarian Tumor (Krukenberg)		-Both ovaries affected -Symptoms of 1 ^o tumor site -Pain referred to abdomen -Irregular bleeding -Painful intercourse	-Death typical in <1 year -Rupture and seed peritoneum	-Signet ring cells (mucin filled) -Stain: PAS -Tumor marker: CA 125 -E-cadherin mutation -Pseudomyxoma	-Spread from 1 of 3 primary sites: stomach, colon, or breast. - Differentiate from mucin tumor of ovary because those are unilateral, not bilateral.
Mature Cystic Teratoma		-Asymptomatic usually -Calcified tooth on X-ray -Can feel dragging in ab. -Mass can be painful	-Infertility -Rupture -Torsion -Malignant transformation ^Squamous cell carcinoma -Keratin pearls then present	-Teeth, hair, and sebaceous secretions. -Females: Benign dermoid cyst - Totipotent germ cell origin -Exophytic growth (outward)	-Derived from 2+ germ layers and occurs in ovaries and testes. - Differentiate between mixed tumor because those are from 1 layer that makes 2 different cells types, teratoma is >2 layers.
Wilm's Tumor		-Children 2-5 yo -Family history (bilateral) -Palpable abdominal mass -Abdominal pain -Hypertension	-Early metastasis to liver and lungs	-Divergent differentiation -Mutation in WT1 & WT2 (TSG) -Nephrogenic nests -Triphasic: -Blastemal ($\uparrow$ nuclear:cyto) -Stromal/mesenchymal (spindle-shaped cells) -Epithelial (def. tubules/glomeruli)	- Malignant mixed tumor. -WAGR tumor has a 33% change of developing into Wilm's. -Sporadic type tend to be unilateral. - Differentiate from benign mixed tumor because no mobile mass.

Neoplasms	Slides	Presentation	Complication	Key words	Areas affected/other
Retinoblastoma		-Children ~2 yo -Family history (bilateral) -Visual impairment -Leukocoria	-Familial: osteosarcoma later -Removal of eyes: blindness	-Cat's eye reflex -Flexner-Wintersteiner rosettes -Mutation in Rb gene -Familial: 1 hit at birth -Sporadic: None at birth ^Unilateral	-Neoplastic tumor of the eye. -Differentiate from neuroblastoma because we have true rosettes, where as neuroblastoma has pseudorosettes (not around central lumen).

-Tissue Biopsy all cancers for diagnosis! Carcinoma (epithelial origin): cytokeratin stain, Sarcoma (mesenchymal origin): Vimentin stain, Muscle origin: Desmin stain.

-Centennial node is the first lymph node that has received metastasis.

-Lymphocytes don't have a benign form of cancer.

-No chance of malignancy: lipoma, capillary hemangioma, and leiomyoma.

-Small chance of malignancy: salivary gland adenoma (mixed tumor) and benign dermoid cyst (teratoma).

-100% chance of malignancy: FAP, HNPCC, osteosarcoma, Wilm's tumor (mixed tumor), and retinoblastoma.